

Exercise 1.

$$\frac{\frac{}{A, A \vdash A} \text{ax}}{\vdash A \Rightarrow A \Rightarrow A} \Rightarrow \text{I} \times 2$$

$$\frac{\frac{\frac{}{(A \Rightarrow A) \Rightarrow B} \Rightarrow \text{I} \text{ ax}}{(A \Rightarrow A) \Rightarrow B} \Rightarrow \text{E} \text{ ax}}{\vdash ((A \Rightarrow A) \Rightarrow B) \Rightarrow B} \Rightarrow \text{I}$$

Exercise 2.

$$\frac{\frac{\frac{}{\neg(A \vee B), A \vdash A \vee B} \vee \text{I} \text{ ax}}{\neg(A \vee B), A \vdash \perp} \Rightarrow \text{E} \text{ ax} \quad \frac{\frac{}{\neg(A \vee B), B \vdash A \vee B} \vee \text{I} \text{ ax}}{\neg(A \vee B), B \vdash \perp} \Rightarrow \text{E} \text{ ax}}{\vdash \neg(A \vee B) \Rightarrow \neg A \wedge \neg B} \Rightarrow \text{I} + \wedge \text{I} + \neg \text{I}$$

$$\frac{\frac{\frac{}{\Gamma, \neg A \vdash A} \neg \text{E} \text{ ax}}{\Gamma, \neg A \vdash \perp} \Rightarrow \text{E} \text{ ax} \quad \frac{\frac{}{\Gamma, \neg B \vdash B} \neg \text{E} \text{ ax}}{\Gamma, \neg B \vdash \perp} \Rightarrow \text{E} \text{ ax}}{\Gamma \vdash \neg A \vee \neg B, A \wedge B \vdash \perp} \vee \text{E} \text{ ax} \Rightarrow \text{I}$$

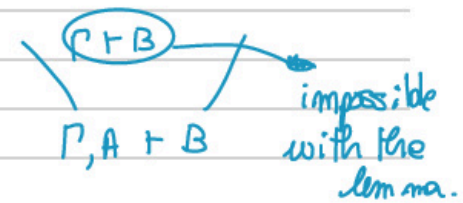
Exercise 3.

$\Rightarrow \text{I}$ is reversible, $\Rightarrow \text{E}$ is not
 $\wedge \text{I}$ is reversible, $\neg \text{E}$ is not
 $\vee \text{I}$, $\vee \text{E}$ are not reversible

$\left(\frac{\vdash T}{\vdash \perp} \right)$ would be admissible with $A = \perp$ and $B = T$
 $\left(\frac{\vdash T}{\vdash T \wedge \perp} \right)$ would be admissible
 $\left(\frac{\vdash T \vee \perp}{\vdash \perp} \right)$ and $\left(\frac{\vdash T}{\vdash \perp \vee \perp} \right)$ would be admissible

Exercise 4. For any proof of $\Gamma \vdash A$, all sequents are of the form $\Gamma, \Delta \vdash A'$ (by induction).

Therefore, if weakening was derivable, then
 Similarly with exchange.



Exercise 5.

Given π a proof tree of $\Gamma, A, B \vdash C$, we proceed by induction on π to show that $\Gamma, A \wedge B \vdash C$ is provable.

$$\frac{\frac{}{\Gamma, A \wedge B, \Delta \vdash A \wedge B} \text{ax}}{\Gamma, A \wedge B, \Delta \vdash A} \wedge \text{E}$$

$$\frac{\frac{}{\Gamma, A \wedge B, \Delta \vdash A \wedge B} \text{ax}}{\Gamma, A \wedge B, \Delta \vdash B} \wedge \text{E}$$

By induction on Γ , the rule $\frac{\Delta \Gamma \vdash A}{\Gamma \vdash A}$ is admissible and reversible
and we conclude by reversibility of $\Rightarrow_i$.

Exercise 6.

Rule $\frac{\Gamma \vdash A \quad \Delta \vdash B}{\Gamma, \Delta \vdash A \wedge B}$ is admissible as, if $\pi: \Gamma \vdash A$
and $\pi': \Delta \vdash B$,
then

$$\frac{\frac{\frac{\pi}{\Gamma \vdash A}}{\Gamma, \Delta \vdash A} \text{weakening}^* \quad \frac{\frac{\frac{\pi'}{\Delta \vdash B}}{\Delta, \Gamma \vdash B} \text{weakening}^* \quad \text{exchange}^*}{\Gamma, \Delta \vdash B}}{\Gamma, \Delta \vdash A \wedge B}$$

Similarly for other rules.

Exercise 7.

$$\frac{\frac{\text{ok}}{\Gamma, A \vdash B} \quad \frac{\text{ok}}{\Gamma, B \vdash A} \quad \wedge_i + \Rightarrow_i}{\Gamma \vdash A \Leftrightarrow B} \quad \frac{\frac{\text{ok}}{\Gamma \vdash A \Leftrightarrow B} \quad \text{ok}}{\Gamma \vdash B \Rightarrow A} \quad \wedge_e \quad \frac{\text{ok}}{\Gamma \vdash A} \quad \Rightarrow_E$$

Exercise 8.

Define $\neg' A := A \Rightarrow \perp$ and $\top' := \neg' \perp = \perp \Rightarrow \perp$.

Exercise 9.

$$\text{If } \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash A} \text{ , } \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash B} \text{ and } \frac{\Gamma \vdash A \quad \Gamma \vdash B}{\Gamma \vdash A \wedge B}$$

$$\text{then } \frac{\frac{A \wedge B \vdash A \wedge B}{A \wedge B \vdash A} \text{ ax } \quad \text{same}}{A \wedge B \vdash A \wedge B} \quad \frac{\frac{A \wedge B \vdash A \wedge B}{A \wedge B \vdash A} \text{ ax } \quad \text{same}}{A \wedge B \vdash A \wedge B} \quad \wedge_i + \Rightarrow_i$$

$$\vdash A \wedge B \Leftrightarrow A \wedge B$$

We can proceed similarly for other connectives.

Exercise 10.

$$\text{If } \Gamma, A \vdash B \text{ then } \frac{\frac{\frac{\Gamma, A \vdash B}{\Gamma, \neg B, A \vdash B}}{\Gamma, \neg B, A \vdash \perp}}{\Gamma, \neg B \vdash \neg A}$$

Exercise 11.

(*)
We need that $\vdash \neg R$ is provable. Then,

$(A \text{ implies } R \text{ which implies } \perp) \text{ implies } \neg A$
(*)

$\neg A \text{ implies } (A \text{ implies } \perp \text{ implies } R)$
ex falso.

It is necessary as $\vdash R \Rightarrow R$ is true and thus so is $\vdash \neg R$.

Exercise 12.

If for all A , $\vdash A$ is provable then so is $\vdash \perp$, and $\vdash A \wedge \neg A$.

If $\vdash \perp$ is provable then so is $\vdash A$ for all A (by $\perp_e$)

If $\vdash A \wedge \neg A$ then $\neg A$ by $\neg e$ and $\neg e$.

Exercise 13.

By induction on M .

- if $\Gamma \vdash x : A$ then necessarily x occurs in Γ .

$$\Gamma' \vdash x : A'$$

as we work with Church style, $A = A'$ and $FV(M) = \{x\} \subseteq \Gamma'$

Conversely, if $x \in \Gamma'$ and $A = A'$ then $\Gamma' \vdash x : A$

- if $\Gamma \vdash \lambda x. M : A \Rightarrow B$ then necessarily $\Gamma, x : A \vdash M : B$
if $\Gamma' \vdash \lambda x. M : A' \Rightarrow B'$ then necessarily $\Gamma', x : A' \vdash M : B'$

by Church style, $A = A'$. by ih, $B = B'$ and $FV(M) \subseteq \Gamma', x$
thus $FV(\lambda x. M) \subseteq \Gamma'$

conversely, it works

- all other cases work.

Exercise 14

$\lambda u.v.$ case $v \{ im_1 a. pr_1 u a \mid im_2 b. pr_2 u b \}$.

$\lambda x.y.x$ and $\lambda f.f(\lambda x.x)$.

There are 2 terms w/o destructors of the type $X \Rightarrow X \Rightarrow X$.
Similarly without elim rules, there are 2 derivations of $\vdash X \Rightarrow X \Rightarrow X$.

Exercise 15.

$$\begin{aligned}
 x &\xrightarrow{R_1 R_2} y \quad \# \exists C[_], u, v, \quad \begin{array}{l} x = C[u] \\ y = C[v] \\ u R_1 \cup R_2 v \end{array} \\
 &\quad \# \exists C[_], u, v, \quad \begin{array}{l} x = C[u] \\ y = C[v] \\ u R_1 v \end{array} \quad \vee \quad \exists C[_], u, v, \quad \begin{array}{l} x = C[u] \\ y = C[v] \\ u R_2 v \end{array} \\
 &\quad \# \quad x \xrightarrow{R_1} y \quad \vee \quad x \xrightarrow{R_2} y
 \end{aligned}$$

If $x R_1 y R_1 z$ and $z R_2 w$ then using context $[_]$,

$$x \xrightarrow{R_1 R_2} w \quad \text{but not} \quad x \xrightarrow{R_1} w \quad \text{nor} \quad x \xrightarrow{R_2} w$$

Exercise 16.

Define $\overline{x \Rightarrow z}$

$$\begin{aligned}
 &\frac{M \Rightarrow M' \quad N \Rightarrow N'}{(M, N) \Rightarrow (M', N')} & \frac{M \Rightarrow M'}{\lambda x. M \Rightarrow \lambda x. M'} & \frac{M \Rightarrow M'}{pr_i M \Rightarrow pr_i M'} \\
 &\frac{M \Rightarrow M' \quad N \Rightarrow N'}{MN \Rightarrow M'N'} & \frac{M \Rightarrow M'}{\text{case } M \{ \} \Rightarrow \text{case } M' \{ \}} & \frac{M \Rightarrow M'}{im_i M \Rightarrow im_i M'} \\
 &\frac{M \Rightarrow M' \quad N \Rightarrow N' \quad P \Rightarrow P'}{\left(\begin{array}{c} \text{case } M \{ \\ im_1 u. N \\ im_2 v. P \end{array} \right) \Rightarrow \left(\begin{array}{c} \text{case } M' \{ \\ im_1 u. N' \\ im_2 v. P \end{array} \right)} & \frac{M \Rightarrow M' \quad M_2 \Rightarrow M'_2 \quad M_3 \Rightarrow M'_3}{\text{case } (im_i M) \{ im_1 u. M_2 \mid im_2 v. M_3 \} \Rightarrow N'_i [M'/u]} & \\
 &\frac{M_1 \Rightarrow M'_1 \quad M_2 \Rightarrow M'_2}{pr_i (M_1, M_2) \Rightarrow M'_i} & \frac{M \Rightarrow M' \quad N \Rightarrow N'}{(\lambda x. M) N \Rightarrow M' [N'/x]} &
 \end{aligned}$$

We have that $(\Rightarrow)^* = \beta^*$ and that $\Rightarrow$ has the diamond

property
(lemma: if $M \Rightarrow N'$ and $N \Rightarrow N'$ then $M[N'/x] \Rightarrow M'[N'/x]$.)

Exercise 17.

None of the β rules enjoy subject expansion.

Consider $\omega := \lambda x. x x$
which is untypable
in $ST\lambda C$.
and $I := \lambda x. x$
which has type
 $A \Rightarrow A$

$(\lambda u. I) \omega \beta I$

$\mu_1(I, \omega) \beta I$

$\mu_2(\omega, I) \beta I$

$\text{case } \text{in}_i \omega \{ \text{in}_1 u. I \mid \text{in}_2 v. I \} \beta I.$

Exercise 18.

We have $u : \perp, v : A \vdash (\text{case}^{A \Rightarrow B} u \{ \}) v : B$
and $u : A \vee B, v : C \vdash (\text{case } u \{ \text{in}_1 a. I \mid \text{in}_2 b. I \}) v : C.$
 $\hookrightarrow C \Rightarrow C.$

Exercise 19.

Normal forms

$N ::= \overbrace{x N_1 \dots N_n}^{n \geq 0}$
| $\lambda x. N$
| (N, N')
| $(\mu x_i N) \overbrace{N_1 \dots N_n}^{n \geq 0}$ if N is not $(-, -)$

Exercise 20.

By induction on N' , let us show if $\Gamma, x : B \vdash N' : A$ and $\Gamma \vdash N_2 : B$ then
 $\omega_m(N'[N_2/x]) \leq \omega(B).$

Exercise 21.

This is the commutation of two elimination rules for different constructors, sometimes removing one of the instances.

Not all elimination rules should commute, however: eg.

$$(pr_1 M)N \not\equiv pr_1 (MN)$$

as M is a pair and not a function.

Exercise 22.

Continuing from exercise 19, we use

$$U ::= N \mid \text{case } N \{ ? \} \mid \text{case } N \{ im_1 u. U \mid im_2 v. U' \}$$

where N spans the grammar from exercise 19.

Exercise 23.

Proof $\pi = \frac{\frac{\overline{\vdash T}}{\vdash Tv \perp} \quad \frac{\overline{\overline{T \vdash T}}}{T \vdash Tv \perp} \quad \frac{\overline{\overline{\perp \vdash \perp}}}{\perp \vdash Tv \perp}}{\vdash Tv \perp}$ has the sub-formula property but does not end with an intro v rule.

Consider $GH(\pi) = \text{case}(im_1 ?) \{ im_1 u. im_1 u \mid im_2 v. im_2 v \}$.

it has the subformula property but it is not a $\beta\delta$ -normal form

$$GH(\pi) \xrightarrow{\beta\delta} im_2 ?.$$

Exercise 24.

Let us show that $(\Box)$ breaks when considering contexts with $\perp$ on v .

Consider, for example,

$$\frac{\overline{AvB \vdash AvB} \quad \frac{\overline{AvB, A \vdash A}}{AvB, A \vdash B \vee A} \quad \frac{\overline{AvB, B \vdash B}}{AvB, B \vdash B \vee A}}{AvB \vdash B \vee A} \quad \text{or} \quad \frac{\overline{\perp \vdash \perp}}{\perp \vdash A}.$$

For a general A, B (e.g. a variable), there is no proof

of the sequents $A \vee B \vdash B \vee A$ and $\perp \vdash A$ that end with an intro rule.

By induction on π a proof of $\Gamma \vdash A$.

For introduction rules, we have no work to do.

It remains three cases

- $\frac{\Gamma \vdash A_1 \wedge A_2}{\Gamma \vdash A_i}$ if there is (by ih) a proof of $\Gamma \vdash A_1 \wedge A_2$ that ends in an intro rule we extract that $\Gamma \vdash A_i$ proof.

otherwise $A_1 \wedge A_2$ subformula of Γ and thus so does A_i .

- $\frac{\Gamma \vdash A \Rightarrow B \quad \Gamma \vdash A}{\Gamma \vdash B}$ if $\Gamma \vdash A \Rightarrow B$ has a proof that ends w/ intro, then we have a proof of $\Gamma, A \vdash B$ and it must either end in intro (ok) or subformula of Γ, A .

The only interesting case is if B subf. of A and we apply ih to $\Gamma \vdash A$.

otherwise that's it.

- $\overline{\Gamma, A \vdash A}$ A subformula of Γ, A

Exercise 25.

Consider $A = X$. If $\vdash X \vee \neg X$ was provable in NJ then either $\vdash X$ or $\vdash \neg X$.

We substitute X by $\perp / \top$ to obtain $\vdash \perp / \vdash \top$

and thus NJ is inconsistent. Contradiction.

Exercise 26.

DNE $\Rightarrow$ EM

$$\begin{array}{c}
 \frac{\frac{\frac{\frac{\frac{\Gamma, EM, A \vdash A}{\Gamma, EM, A \vdash \neg EM}}{\Gamma, EM, A \vdash \perp}}{\Gamma, EM \vdash \neg A}}{\Gamma, EM \vdash EM}}{\Gamma, EM \vdash \perp} \\
 \frac{\Gamma, EM \vdash \perp}{\Gamma \vdash \neg \neg EM} \quad \Gamma \vdash DNE \\
 \hline
 \Gamma \vdash EM
 \end{array}$$

CP $\Rightarrow$ DNE

$$\begin{array}{c}
 \frac{\frac{\frac{\dots \vdash \neg A}{\neg A, \neg \neg A \vdash \perp}}{\neg A \vdash \neg \neg A}}{\vdash \neg A \Rightarrow \neg \neg A} \quad \vdash CP \\
 \hline
 \vdash DNE
 \end{array}$$

DNE $\Rightarrow$ PL

$$\begin{array}{c}
 \frac{\frac{\frac{\frac{\frac{\Gamma, A \vdash A}{\Gamma \vdash A \Rightarrow B} \quad \Gamma, A \vdash \neg A}{\Gamma \vdash A \Rightarrow \neg B}}{\Gamma \vdash (A \Rightarrow B) \Rightarrow A, \neg A \vdash \perp}}{\dots \vdash DNE} \quad (A \Rightarrow B) \Rightarrow A \vdash \neg \neg A \\
 \frac{(A \Rightarrow B) \Rightarrow A \vdash A}{\vdash ((A \Rightarrow B) \Rightarrow A) \Rightarrow A}
 \end{array}$$

EM $\Rightarrow$ CP

$$\begin{array}{c}
 \frac{\frac{\frac{\frac{\Gamma, \neg A \vdash \neg A \Rightarrow \neg B}{\Gamma, \neg A \vdash \neg B}}{\Gamma, \neg A \vdash \perp}}{\Gamma, \neg A \vdash A} \quad \vdash CP \\
 \frac{\Gamma \vdash A \vee \neg A \quad \Gamma, A \vdash A \quad \Gamma, \neg A \vdash A}{\Gamma \vdash \neg A \Rightarrow \neg B, B \vdash A} \Rightarrow; \times 2 \\
 \hline
 \vdash CP
 \end{array}$$

PL $\Rightarrow$ DNE

$$\begin{array}{c}
 \frac{\frac{\frac{\dots \vdash \neg \neg A}{\neg \neg A, \neg A \vdash \perp}}{\neg \neg A, \neg A \vdash A}}{\neg \neg A \vdash \neg A \Rightarrow A} \\
 \frac{\neg \neg A \vdash PL \quad \neg A \vdash A}{\vdash DNE}
 \end{array}$$

Exercise 27.

As all of the four axiom schemas from exercise 26 are equivalent, we only need to show one of them:

$$\begin{array}{c}
 \frac{\frac{\frac{\dots \vdash \neg A}{\neg \neg A, \neg A \vdash \perp}}{\neg \neg A \vdash A} \quad \neg \neg A \\
 \hline
 \vdash DNE
 \end{array}$$

Exercise 28.

By exercise 32, we only need to show one of them:

$$\begin{array}{c}
 \frac{\frac{\frac{A \vdash A, \perp}{A \vdash \perp, A}}{\vdash \neg A, A}}{\vdash A, \neg A} \\
 \hline
 \vdash A \vee \neg A
 \end{array}$$

Exercise 29.

$$\begin{array}{c}
 \frac{\frac{\frac{\vdash (ex 28)}{\Gamma \vdash A \vee \neg A} \quad \Gamma, A \vdash A \quad \Gamma, \neg A \vdash \perp}{\Gamma, \neg A \vdash A}}{\Gamma \vdash A} \quad \text{by ex 32.}
 \end{array}$$

Exercise 30.

$$\begin{array}{c}
 \frac{\frac{\frac{\dots \vdash A \quad \dots \vdash B}{\dots \vdash A \wedge B} \quad \dots \vdash \neg(\theta \wedge \theta)}{\neg(A \wedge B), A, B \vdash \perp, \perp} \quad \frac{\Gamma, \neg A \vdash A \wedge B}{\Gamma, \neg A \vdash A} \quad \text{similar} \\
 \frac{\neg(A \wedge B), A \vdash \neg B, \perp}{\neg(A \wedge B), A \vdash \perp, \neg B} \quad \frac{\Gamma \vdash \neg A \vee \neg B \quad \Gamma, \neg A \vdash \perp \quad \Gamma, \neg B \vdash \perp}{\Gamma \vdash \neg A \vee \neg B, A \wedge B \vdash \perp} \quad \text{ex 32} \\
 \frac{\neg(A \wedge B) \vdash \neg A, \neg B}{\neg(A \wedge B) \vdash \neg A \vee \neg B} \quad \frac{\Gamma \vdash \neg A \vee \neg B, A \wedge B \vdash \perp}{\neg A \vee \neg B \vdash \neg(A \wedge B)} \\
 \hline
 \vdash \neg(A \wedge B) \Leftrightarrow \neg A \vee \neg B
 \end{array}$$

Exercise 31. Given a proof π of $\Gamma \vdash \Delta, \Theta$, we proceed by induction on π . Most cases are uninteresting, if we suppose Δ is non-empty.

- $\wedge$ intro if Δ is A and $\Theta = B, \Theta'$

$$\begin{array}{c}
 \text{by ih.} \\
 \frac{\vdash A \vee B, C, \Theta'}{\vdash A, B, C, \Theta'} \\
 \frac{\vdash A, C, B, \Theta'}{\vdash A, C, B, \Theta'} \\
 \underbrace{\quad}_{\Delta} \quad \underbrace{\quad}_{\Theta}
 \end{array}$$

- $\wedge$ elim for a similar case also requires an exchange.
- exchange (we might need to exchange the 1st with the $(i+1)$ th)

If we have Δ empty, necessarily, Θ is non-empty and we can recover the assumption of Δ non-empty with an exchange.

Exercise 32.

Most of the rules exists in NC, except $\wedge$ elim and $\wedge$ intro

$$\begin{array}{c}
 \frac{\Gamma, B \vdash C}{\Gamma \vdash B \Rightarrow C} \quad \frac{\frac{\Gamma, A \vdash C}{\Gamma \vdash A \Rightarrow C} \quad \Gamma \vdash A \Rightarrow C, C}{\Gamma \vdash C, B} \quad \frac{\Gamma \vdash A \vee B}{\Gamma \vdash A, B} \quad \frac{\frac{\vdash A_i}{\vdash A_1, A_2} \quad \text{weakening}}{\vdash A_1 \vee A_2} \\
 \hline
 \frac{\Gamma \vdash C, C}{\Gamma \vdash C}
 \end{array}$$

Exercise 33.

$\Gamma \vdash \Delta$ iff $\Gamma \vdash \vee \Delta$ by repeated on intro (reversible)
 iff $\bigwedge \Gamma \vdash \vee \Delta$ as rules $\frac{\Gamma, A \wedge B \vdash \Delta}{\Gamma, A, B \vdash \Delta}$ and $\frac{\Gamma, A, B \vdash \Delta}{\Gamma, A \wedge B \vdash \Delta}$ are
 admissible (similar to ex 5)
 iff $\vdash \bigwedge \Gamma \Rightarrow \vee \Delta$ with $\Rightarrow_I$ (reversible).

Exercise 34.

Rules $\vee_i, \vee_e, \Rightarrow_i, \neg_i, \neg_e, \wedge_i$ are reversible

Exercise 35.

$$\frac{\frac{\frac{\dots \vdash A \quad \dots \vdash A}{\dots \vdash \perp}}{\dots \vdash A \vee B} \quad \frac{\dots \vdash A \vdash B \quad \dots \vdash B}{A \vee B, \neg A \vdash B} \quad \text{ex 28} \quad \frac{\dots \vdash A \vee B}{A \vee B, \neg A \vdash B} \quad \frac{\dots \vdash A \vee B \quad \dots \vdash B, A}{\dots \vdash A, B} \quad \frac{\dots \vdash A, B}{\dots \vdash A \vee B} \quad \frac{\dots \vdash A \vee B}{\dots \vdash A \vee B}$$

$$\frac{A \vee B, \neg A \vdash B}{A \vee B \vdash \neg A \Rightarrow B} \quad \frac{\neg A \Rightarrow B \vdash A \vee B}{\vdash (A \vee B) \Leftrightarrow (\neg A \Rightarrow B)}$$

$$\frac{\frac{\dots \vdash A \wedge B}{\dots \vdash B} \quad \frac{\dots \vdash A \Rightarrow B \quad \dots \vdash A}{A \wedge B, A \Rightarrow B \vdash \perp} \quad \frac{\dots \vdash A \wedge B}{\dots \vdash A \wedge B} \quad \frac{\dots \vdash A \wedge B}{\dots \vdash A \wedge B} \quad \frac{\dots \vdash A \wedge B}{\dots \vdash A \wedge B} \quad \frac{\dots \vdash A \wedge B}{\dots \vdash A \wedge B}$$

$$\frac{A \wedge B, A \Rightarrow B \vdash \perp}{A \wedge B \vdash \neg A \Rightarrow \neg B} \quad \frac{\dots \vdash A \wedge B}{\dots \vdash A \wedge B} \quad \frac{\dots \vdash A \wedge B}{\dots \vdash A \wedge B} \quad \frac{\dots \vdash A \wedge B}{\dots \vdash A \wedge B} \quad \frac{\dots \vdash A \wedge B}{\dots \vdash A \wedge B}$$

$$\vdash (A \wedge B) \Leftrightarrow \neg (A \Rightarrow \neg B)$$

In NJ, we can show $\vdash \neg A \Rightarrow \neg A$ and we could thus prove EM.
 Similarly, we know $\vdash A \Rightarrow \neg \neg A$ and we could thus prove $\vdash \neg (\neg \neg A)$
 and thus DNE.

Both cases lead to a contradiction.

Exercise 36.

$$\begin{array}{c} \frac{\dots \vdash A \quad \dots \vdash \neg A}{\dots \vdash \neg \neg A} \\ \frac{\dots \vdash \neg \neg A}{\dots \vdash \neg \neg A} \\ \frac{\neg \neg A, A \vdash \perp}{\vdash \neg \neg A \Rightarrow \neg A} \end{array} \quad \begin{array}{c} \frac{\dots \vdash \neg A \quad \dots \vdash \neg A}{\neg A, \neg A \vdash \perp} \\ \frac{\neg A, \neg A \vdash \perp}{\vdash \neg A \Rightarrow \neg \neg A} \end{array} \quad \neg$$

$$\vdash \neg A \Leftrightarrow \neg \neg A$$

Exercise 37.

From an NC proof of $\Gamma \vdash A \wedge B$ we can always deduce a proof of $\Gamma \vdash A$ and $\Gamma \vdash B$.

However, we can't always deduce a proof of $\Gamma \vdash A$ or $\Gamma \vdash B$ given a proof of $\Gamma \vdash A \vee B$. (eg EN).

For $\neg A$ we use ex 36.

Exercise 38.

Lemma 1: $\neg \neg A^G \vdash A^G$ in NS $\forall A$

Lemma 2: $A^G \vdash \neg \neg A^G$ in NS $\forall A$.

Lemma 3: if $\Gamma^G \vdash A^G$ in NS then $\Gamma \vdash A$ in NC

Lemma 4: the converse of lemma 3.

These lemmas follow by induction (very long)

Exercise 39.

$\underbrace{C[\underbrace{\underbrace{B}_{\mathcal{C}}]}_{\mathcal{N}}]}_A \rightarrow \underbrace{N[\underbrace{\underbrace{A \Rightarrow B}_{\mathcal{C}}}]_B}_{A \Rightarrow B}$ thus $\mathcal{C}: (A \Rightarrow B) \Rightarrow B \Rightarrow A$

and $\underbrace{C[\underbrace{\underbrace{A}_{\mathcal{C}}}]_A}_{A} \rightarrow \underbrace{C[\underbrace{N[\underbrace{\underbrace{A \Rightarrow B}_{\mathcal{C}}}]_B}_{(A \Rightarrow B) \Rightarrow A}]}_{A \Rightarrow B}$ thus $\mathcal{C}: ((A \Rightarrow B) \Rightarrow A) \Rightarrow A$.

Exercise 40.

Consider u and v two λ -terms. Let us prove that $u =_{\alpha\beta} v$ where $x \notin \text{FV}(u, v)$.

Consider the term

$$(\lambda x. u) (\mathcal{C}(\lambda x. v)) \xrightarrow{\beta} u$$

$$\begin{array}{c} \mathcal{C} \downarrow \text{(with } C[\] = (\lambda x. u)[\] \text{)} \\ (\lambda x. v) (\lambda y. (\lambda x. u) y) \xrightarrow{\beta} v \end{array}$$

Exercise 41.

Lemma: if F, F' are $\mathcal{E}$ -contexts and $R, P \in \text{dom } \beta \cup \text{dom } \mathcal{C} \cup \text{dom } \alpha$ such that $F[R] = F'[P]$ then $R = P$ and $F = F'$.

The results follow from the lemma.

Exercise 42. We do the exercise assuming $B = \perp$. We can use LE to change for an arbitrary B.

Define $cc' := \lambda f. \lambda g. \mathcal{C}(\lambda g. g(f\ g))$.

$$\begin{aligned} E[cc' M] &\longrightarrow_{\beta} E[\mathcal{C}(\lambda g. g(M\ g))] \xrightarrow{\mathcal{C}^{CBN}} (\lambda g. g(M\ g))(\lambda y. E[y]). \\ &\longrightarrow_{\beta} (\lambda y. E[y])(\lambda y. E[y]) \\ &\longrightarrow_{\beta} E[M(\lambda y. E[y])]. \end{aligned}$$

Define $\mathcal{C}' := \lambda f. \lambda g. \mathcal{C}(\lambda g. \lambda f. g)$.

$$\begin{aligned} E[\mathcal{C}' M] &\longrightarrow E[cc(\lambda g. \lambda f. g)] \xrightarrow{*} E[\lambda f. f(M(\lambda y. E[y]))] \\ &\longrightarrow M(\lambda y. E[y]) \end{aligned}$$

Exercise 43.

By induction on M, similar to ex 13.

Exercise 44.

We have

$$\begin{array}{c} \frac{}{f: \neg\neg A, x:A \vdash x:A} \\ \frac{}{f: \neg A, x:A \vdash [\alpha] x : \perp \mid \alpha:A} \\ \hline \frac{}{f: \neg A \vdash f.m \lambda \alpha. \lambda x. [\alpha] x : \neg A \mid \alpha:A} \\ \hline \frac{}{f: \neg A \vdash f(\lambda x. [\alpha] x) : \perp \mid \alpha:A} \\ \frac{}{f: \neg A \vdash \mu \alpha. f(\lambda x. [\alpha] x) : A} \\ \hline \vdash \lambda f. \mu \alpha. f(\lambda x. [\alpha] x) : \neg A \Rightarrow A \end{array}$$

and that proof enjoys the sub-formula property

Exercise 45.

We have

$$\begin{array}{c} \frac{}{\dots \vdash x : A \mid \beta:B} \\ \frac{}{\dots \vdash [\alpha'] x : \perp \mid \alpha':B, \beta:B} \\ \hline \frac{}{f: \dots, x:A \vdash \mu \beta. [\alpha'] x : B \mid \alpha':B} \\ \hline \frac{}{f: \dots \vdash f(\lambda x. \mu \beta. [\alpha'] x) : B \mid \alpha':B} \\ \hline \frac{}{f: \dots \vdash [\alpha] f(\lambda x. \mu \beta. [\alpha'] x) : \perp \mid \alpha:B, \alpha':B} \\ \hline \frac{}{f: \dots \vdash [\alpha] f(\lambda x. \mu \beta. [\alpha'] x) : \perp \mid \alpha:B} \text{ contr} \quad \Delta \\ \hline \frac{}{f: (A \Rightarrow B) \Rightarrow A \vdash \mu \alpha [\alpha] f(\lambda x. \mu \beta. [\alpha'] x) : B} \\ \hline \vdash \lambda f. \mu \alpha [\alpha] f(\lambda x. \mu \beta. [\alpha'] x) : ((A \Rightarrow B) \Rightarrow A) \Rightarrow B \end{array}$$

and that proof enjoys the sub-formula property

Exercise 66.

If $\Gamma, x:B \vdash M:A \mid \Delta$ and $\Gamma \vdash N:B \mid \Delta$ then $\Gamma \vdash M[N/x]:A \mid \Delta$.

If $\Gamma \vdash M:A \mid \alpha:B, \Delta$ then $\Gamma \vdash M[\rho/\alpha]:A \mid \rho:B, \Delta$
with $\alpha, \rho \notin \text{dom } \Delta$.

If $\Gamma \vdash M:\perp \mid \alpha:A \Rightarrow B, \Delta$ and $\Gamma \vdash N:A \mid \Delta$

then $\Gamma \vdash M[\ulcorner \alpha \urcorner_{\perp} \ulcorner N \urcorner_{\perp} / \ulcorner \alpha \urcorner_{\perp}]:\perp \mid \alpha:B, \Delta$

with $\alpha \notin \text{dom } \Delta$

Exercise 67.

By induction on M , we have 5 cases

- if $\Gamma \vdash x:A \mid \Delta$ then $\frac{\dots \vdash k:\neg A^* \quad \dots \vdash x:\neg A^*}{\dots, k:\neg A^* \vdash xk : R}$
 $\neg\neg \Gamma^*, \neg \Delta^* \vdash \lambda k. xk : A^{K_0}$

so $x:\neg A^*$ occurs in $\neg\neg \Gamma^*$

- if $\Gamma \vdash MN:A \mid \Delta$ then $\Gamma \vdash M:B \Rightarrow A \mid \Delta$
and $\Gamma \vdash N:B \mid \Delta$
thus $\neg\neg \Gamma^*, \neg \Delta^* \vdash M^{K_0}:\neg\neg (B^{K_0} \Rightarrow A^{K_0})$
and $\neg\neg \Gamma^*, \neg \Delta^* \vdash N^{K_0}:B^{K_0}$

$$\frac{\vdots \quad \frac{\dots \vdash m:B^{K_0} \Rightarrow A^{K_0} \quad \dots \vdash N^{K_0}:B^{K_0} \quad \dots \vdash k:\neg A^*}{\dots \vdash m:B^{K_0} \Rightarrow A^{K_0} \quad \dots \vdash N^{K_0}:B^{K_0} \quad \dots \vdash k:\neg A^*} \text{wk}}{\neg\neg \Gamma^*, \neg \Delta^*, k:\neg A^* \vdash M^{K_0}:\neg\neg (B^{K_0} \Rightarrow A^{K_0})} \text{wk} \quad \frac{\neg\neg \Gamma^*, \neg \Delta^*, k:\neg A^*, m:B^{K_0} \Rightarrow A^{K_0} \vdash mN^{K_0} : k:B}{\neg\neg \Gamma^*, \neg \Delta^*, k:\neg A^* \vdash \lambda m. mN^{K_0} : k:\neg (B^{K_0} \Rightarrow A^{K_0})}$$

$$\frac{\neg\neg \Gamma^*, \neg \Delta^*, k:\neg A^* \vdash M^{K_0}(\lambda m. mN^{K_0} k) : R}{\neg\neg \Gamma^*, \neg \Delta^* \vdash \lambda k. M^{K_0}(\lambda m. mN^{K_0} k) : \neg\neg A^*}$$

- if $\Gamma \vdash \lambda x. M : A \Rightarrow B \mid \Delta$ then $\Gamma, x:A \vdash M:B \mid \Delta$
thus $\neg\neg \Gamma^*, x:\neg A^*, \neg \Delta^* \vdash M^{K_0}:B^{K_0}$

$$\frac{\vdots \quad \frac{\neg\neg \Gamma^*, x:\neg A^*, \neg \Delta^* \vdash M^{K_0}:B^{K_0}}{\neg\neg \Gamma^*, \neg \Delta^* \vdash \lambda x. M^{K_0} : A^{K_0} \Rightarrow B^{K_0}}}{\dots \vdash k:\neg (A^{K_0} \Rightarrow B^{K_0})} \quad \frac{\neg\neg \Gamma^*, \neg \Delta^* \vdash \lambda x. M^{K_0} : A^{K_0} \Rightarrow B^{K_0}}{\neg\neg \Gamma^*, \neg \Delta^*, k:\neg (A^{K_0} \Rightarrow B^{K_0}) \vdash k(\lambda x. M^{K_0}) : R}$$

$$\neg\neg \Gamma^*, \neg \Delta^* \vdash \lambda k. k(\lambda x. M^{K_0}) : \neg\neg (A^{K_0} \Rightarrow B^{K_0})$$

• If $\Gamma \vdash [\alpha] M : \perp \mid \Delta_1, \alpha:A, \Delta_2$ then $\Gamma \vdash M : A \mid \Delta_1, \Delta_2$
 Thus $\neg \Gamma^*, \neg \Delta_1^*, \neg \Delta_2^* \vdash M^{K_0} : A^{K_0}$

$$\frac{\vdots \quad \dots \vdash M^{K_0} : \neg A^* \quad \dots \vdash \alpha : \neg A^*}{\neg \Gamma^*, \neg \Delta_1^*, \alpha : \neg A^*, \neg \Delta_2^* \vdash M^{K_0} \alpha : R}$$

• if $\Gamma \vdash \mu \alpha. M : A \mid \Delta_1, \Delta_2$ then $\Gamma \vdash M : \perp \mid \Delta_1, \alpha:A, \Delta_2$
 Thus $\neg \Gamma^*, \neg \Delta_1^*, \alpha : \neg A^*, \neg \Delta_2^* \vdash M^{K_0} : R.$

$$\frac{\vdots \quad \dots, \alpha : \neg A^* \vdash M^{K_0} : R}{\neg \Gamma^*, \neg \Delta_1^*, \neg \Delta_2^* \vdash \lambda \alpha. M^{K_0} : \neg A^*}$$

This concludes the induction.

Exercise 48.

Lemmas about substitution and translation.
 Context closure seems easy

Exercise 49.

- | | | |
|-------------------------|---|---|
| 1. $\vdash$ in NS | 2. $\nvdash$ in NS/NC. | 3. $\lambda y. \lambda x. \lambda f. f y$ |
| 4. $\nvdash$ in NS/NC. | 5. pick $x = \perp$.
$\perp \Rightarrow y$ is possible
in NS | 6. $\nvdash$ in NS
$\vdash$ in NC |
| 7/8. $\nvdash$ in NS/NC | | |

Exercise 50.

$$\frac{\frac{\text{hiv.}}{\vdash \neg \perp}}{\vdash \exists x. \neg x}}{\vdash \neg x}}{\vdash \forall x. \neg x}}{\vdash \neg T}$$

Exercise 51.

We have $\vdash \forall x. A \Rightarrow \neg \exists x. \neg A$ in NS₂
 and $\vdash \exists x. A \Rightarrow \neg \forall x. \neg A$,

but the converse ones are not derivable as, if they were
 and $x \notin FV(A)$, then we would deduce
 $\vdash A \Rightarrow \neg \neg A$.

Exercise 52.

Time consuming but not hard:

for $A \bar{x} B \vdash A * B$ use $x = A * B$

for $A * B \vdash A \bar{x} B$ use intro; elim * and that's about it.

Exercise 53.

Not : Bool $\Rightarrow$ Bool

Not := $\lambda b:\text{Bool}. b \text{ Bool False True}$

Not True $\xrightarrow{P} \text{True Bool False True} \xrightarrow{\beta^*} \text{False}$
 Not False $\xrightarrow{P} \text{False Bool False True} \xrightarrow{\beta^*} \text{True}$

Exercise 54.

We can easily check, for $b_1, b_2 \in \{\text{True}, \text{False}\}$ that

$\text{And}_e \ b_1 \ b_2 =_P \text{True}$ iff $b_1 = b_2 = \text{False}$

otherwise $\text{And}_e \ b_1 \ b_2 =_P \text{False}$.

$\text{And}_s := \lambda b_1 b_2. b_1 \text{ Bool } b_2 \ b_1$.

Exercise 55.

given $M:A_1 \ N:A_2$,

$p_i : A_1 \bar{\wedge} A_2 \Rightarrow A_i$

$\langle M, N \rangle : A_1 \bar{\wedge} A_2$

$\langle M, N \rangle := \lambda X. \lambda f. f \ M \ N$

$p_i := \lambda p. p \ A_i \ (\lambda a_1 a_2. a_i)$.

$p_i \langle M, N \rangle \xrightarrow{\beta^*} (\lambda a_1 a_2. a_i) \ M \ N \xrightarrow{\beta} M_i$

Exercise 56.

$\text{im}_i : A_i \longrightarrow A_1 \bar{\vee} A_2$

$\text{im}_i := \lambda a. \lambda X. \lambda \mu_1 \mu_2. \mu_i \ a$.

$\text{case} : \forall X. A_1 \bar{\vee} A_2 \longrightarrow (A_1 \rightarrow X) \rightarrow (A_2 \rightarrow X) \rightarrow X$.

$\text{case} := \lambda X \lambda p. p \ X$.

Exercise 57.

$\text{succ} : \text{Nat} \rightarrow \text{Nat}$

$\text{succ} := \lambda n. \lambda X. \lambda f x. f (n \ X \ f \ x)$.

$\text{add} : \text{Nat} \rightarrow \text{Nat} \rightarrow \text{Nat}$

$\text{add} := \lambda m \ m'. m \ \text{Nat succ } m'$

$\text{mult} : \text{Nat} \rightarrow \text{Nat} \rightarrow \text{Nat}$
 $\text{mult} := \lambda m m'. m \text{ Nat (add } m')$ 0

$\text{exp} : \text{Nat} \rightarrow \text{Nat} \rightarrow \text{Nat}$
 $\text{exp} := \lambda m m'. m \text{ Nat (mult } m')$ 1.

Exercise 58.

Define $\text{update} : \text{Nat} \times \text{Nat} \rightarrow \text{Nat} \times \text{Nat}$
 $\text{update} := \lambda p. p (\text{Nat} \times \text{Nat}) (\lambda ab. \langle \text{succ } 0, 0 \rangle)$

and $\text{pred} : \text{Nat} \rightarrow \text{Nat}$
 $\text{pred} := \lambda n. p_2 (n (\text{Nat} \times \text{Nat}) \text{ update } \langle 0, 0 \rangle)$

Exercise 59.

All from ex 57 is easy but ex 58 is harder as we would want $\text{pred} : \text{Nat}_{\text{Nat}_x \times \text{Nat}_x} \rightarrow \text{Nat}_{\text{Nat}_x \times \text{Nat}_x}$.

Exercise 60.

Consider a type A . Let us show any $x : A$ is in $\text{CRED}(A)$.

By R_3 , as x is neutral and of type A ,
 and $(\forall M, x \rightarrow M \Rightarrow \dots)$ as x is a normal form,
 thus $x \in \mathcal{U}$

Exercise 61.

$R_1: \text{SN}^A \subseteq \text{SN}$.

R_2 : Assume $n \in \text{SN}^A$ and $M \rightarrow N$ with $N \notin \text{SN}^A$
 thus either $\text{FN} : A$ (impossible by subject reduction)
 or $N \notin \text{SN}$ (impossible by def^o of SN)

R_3 by def of SN^A

Exercise 62.

We want to show $\lambda x. M \in \text{RED}(A \Rightarrow B)_p$ i.e.
 $\forall N \in \text{RED}(A)_p \quad MN \in \text{RED}(B)_p$.

Consider $N \in \text{RED}(A)_p$. As $\text{RED}(B)_p$ is a reducibility
 candidate for $A[B_1, \dots, B_n / x_1, \dots, x_n] =: \bar{A}$,

we apply R3, $(\lambda x M)N$ is neutral and of type B, and use the assumption.

Exercise 63.

By a simple induction on A .

If $A = X \neq Y$ then we have to show $\text{RED}(A \{B/Y\})_\rho = \text{RED}(A)_\rho = \text{RED}(A)_\rho [Y \mapsto \dots]$

If $A = Y$ then $\text{RED}(A \{B/Y\})_\rho = \text{RED}(B)_\rho = \text{RED}(Y)_\rho [Y \mapsto \text{RED}(B)_\rho]$ by def of $\text{RED}(\rightarrow)_\rho$ on vars.

If $A = A' \Rightarrow A''$ then we unfold, substitute by ρ , and refold.

If $A = \forall X. A'$ then similar.

Exercise 64.

By def $=_\beta$ is an equiv relation and context closed

However, it is false that $=_\beta$ is type compatible:

$(\lambda z. \lambda y. y) \Omega$ is not typable yet it is $=_\beta$ to $\lambda z. z$ which is typable.

Exercise 65.

Denote $R := \bigcap_{i \in \mathbb{Z}} R_i$.

we have $M R M$ as $M R_i M$ for every i
same for symmetry & transitivity

If $M R N$ then $\forall i \ M R_i N$, and thus

$\forall C[]$, $C[M] R C[N]$ as $\forall i \ C[M] R_i C[N]$.

If $M R N$ then either I is empty $\leadsto$ exercise doesn't make sense.
or $M R_i N$ for some i
and thus $\Gamma \vdash M : A \quad \Gamma \vdash N : A$ for some Γ, A .

Exercise 66.

We know $=_\beta \subseteq \cong_{\text{Bool}}$. Suppose $M \cong_{\text{Bool}} N$ then, using context $[]$, we have $M =_\beta N$.

Exercise 64.

By parametricity, $\sim_A$ is reflexive.

Let us show by induction on A that

- if $M \sim_A N [\eta: \sigma \leftrightarrow \sigma']$ then $N \sim_A M [\bar{\eta}: \sigma' \leftrightarrow \sigma]$
- if also $N \sim_P [\eta': \sigma' \leftrightarrow \sigma'']$ then $M \sim_{\#P} [\eta \circ \eta': \sigma \leftrightarrow \sigma'']$.

where

$$\bar{\eta}(x) := \{(Q, P) \mid (P, Q) \in \eta(x)\}$$

$$\eta \circ \eta'(x) := \{(P, R) \mid \exists Q, (P, Q) \in \eta(x), (Q, R) \in \eta'(x)\}.$$

Exercise 68.

By induction on A ,

- if $A = X$, then as $M' \sim_X N [\eta: \sigma \leftrightarrow \sigma']$
we have $(M', N) \in \eta(x)$
and by admissibility,

$$M \cong_{\sigma(x)} M' \uparrow_{\eta(x)} N \cong_{\sigma'(x)} N$$

implies $M \sim N [\eta: \sigma \leftrightarrow \sigma']$

- if $A = B \Rightarrow C$, then take $L \sim_B P [\eta: \sigma \leftrightarrow \sigma']$
and we need to prove $M'L \sim_C NP [\eta: \sigma \leftrightarrow \sigma']$.

By hypothesis and context closure, $\square \perp$

$$ML \cong_{\sigma(c)} M'L \sim_B NP [\eta: \sigma \leftrightarrow \sigma']$$

thus by induction hypothesis, we conclude

$$ML \sim_C NP [\eta: \sigma \leftrightarrow \sigma'].$$

- if $A = \forall X, B$, take C, C' and $R: C \leftrightarrow C'$,
we want to show that

$$M'C \sim_B N C' \quad \overbrace{[\eta[x \mapsto C] : \sigma[x \mapsto C] \leftrightarrow \sigma'[x \mapsto C']]}^{\eta'}$$

by hypothesis, $M \cong_{\sigma(\forall X, B)} N$ thus, by context closure ($[C]$)

$$MC \cong_{\sigma(B)[C/x]} M'C \sim_B N C' [\eta']$$

and we conclude by induction hypothesis.

Exercise 69.

Suppose M is a closed term of type $\forall X. X$.

By parametricity, $M \sim_{\forall X. X} M$. Define $\sigma(x) := A = \forall Y. Y$
 $\eta(x) := \emptyset$

Therefore $MA \sim_x MA [\eta : \sigma \leftrightarrow \sigma]$
 contradiction : $(MA, MA) \in \eta(x) = \emptyset$

Exercise 70.

By ext, $M \cong_{\forall X. X \Rightarrow X} \lambda X. \lambda z. z$ iff $\forall A$ closed, $\forall P \cong_A Q$,

$$MAP \cong_A (\lambda X. \lambda z. z) A Q =_B Q$$

By parametricity, $M \sim_{\forall X. X \Rightarrow X} M$.

Define $\sigma(x) := A$ and $\eta(x) := [Q] \stackrel{?}{\cong}_A \rightarrow$ admissible

Therefore $MA \sim_{X \Rightarrow X} MA [\eta : \sigma \leftrightarrow \sigma]$

and thus $MAP \sim_x MAP [\eta : \sigma \leftrightarrow \sigma]$

concluding the exercise: $MAP \cong_A Q$

we have

$$P \sim_x P [\eta : \sigma \leftrightarrow \sigma]$$

$$\text{by } P \cong_A Q \cong_A P.$$

Exercise 71.

First, let us show $M \text{ Bool True False } \cong_{\text{Bool True or}} \cong_{\text{Bool False}}.$

By parametricity, $M \sim_{\text{Bool}} M$

define $\sigma(x) := \text{Bool}$
 $\eta(x) := [\text{True}]_{\sim}^2 \cup [\text{False}]_{\sim}^2$

thus $M \text{ Bool True False} \sim_x M \text{ Bool True False} [\eta: \sigma \leftrightarrow \sigma]$.

as $\begin{cases} \text{True} \sim_x \text{True} [\eta: \sigma \leftrightarrow \sigma] \\ \text{False} \sim_x \text{False} [\eta: \sigma \leftrightarrow \sigma] \end{cases}$

and thus we conclude.

Without loss of generality, suppose $M \text{ Bool True False} \cong_{\text{Bool}} \text{False}$.

Let us show $M \cong_{\text{Bool}} \text{False}$ i.e. by ext $\forall A \text{ closed}, \forall P, Q,$
 $M \text{ A P Q} \cong_A \text{False A P Q}$

By parametricity, $M \sim_{\text{Bool}} M$.

(Re) define $\sigma(x) := A$ $\sigma'(x) := \text{Bool}$
 and $\eta(x) := [P] \times [\text{True}] \cup [Q] \times [\text{False}]$.

We have $M \text{ A P Q} \sim_x M \text{ Bool True False} [\eta: \sigma \leftrightarrow \sigma']$

as $\begin{cases} P \sim_x \text{True} [\eta: \sigma \leftrightarrow \sigma'] \\ Q \sim_x \text{False} [\eta: \sigma \leftrightarrow \sigma'] \end{cases}$
 by def of $\eta(x)$.

Therefore $M \text{ A P Q} \cong_A Q$ as $M \text{ Bool True False} \cong_{\text{Bool}} \text{False}$

Exercise 72.

$\text{And}_1 \cong \text{And}_2$ iff $\forall U, V, \quad \text{And}_1 U V \cong_{\text{And}_2} U V$

By ex 71, $U \cong \text{True}$ or $U \cong \text{False}$ (and same for V).

Consider the case $U \cong \text{True}$ and $V \cong \text{False}$

By context closure and $=_p \subseteq \cong$ we have

$\wedge x. \lambda z y. \text{True } X (\text{False } X z y). y =_p \wedge x \lambda z y. y$
 and $\wedge x. \lambda z y. \text{False } X (\text{True } X z y). y =_p \wedge x \lambda z y. y$.

Therefore $\text{And}_1 U V \cong_{\text{Bool}} \text{And}_2 U V$ in this case

We do the same for all the other cases and conclude.

Exercise 73.

Let us show $M \text{ Nat } S \subseteq \cong_{\text{Nat}} \underline{n}$ for some $n \in \mathbb{N}$.

By parametricity $M \sim_{\text{Nat}} M$ define $\sigma(x) := \text{Nat}$
and $\eta(x) := \bigcup_{n \in \mathbb{N}} [\underline{n}]_{\cong}^c$

thus $M \text{ Nat } S \subseteq \sim_x M \text{ Nat } S \subseteq [\eta : \sigma \mapsto \sigma]$

as $\bullet \subseteq \sim_x \subseteq [\eta : \sigma \mapsto \sigma]$ with $n=0$

$\bullet S \sim_{x \mapsto x} S [\eta : \sigma \mapsto \sigma]$

since, if $U \sim_x U [\eta : \sigma \mapsto \sigma]$ then $U \cong \underline{n}$ for some n

and thus $SU \cong S\underline{n} = \underline{n+1}$

therefore $SU \sim_x SU [\eta : \sigma \mapsto \sigma]$.

Let n such that $M \text{ Nat } S \subseteq \cong \underline{n}$

By ext, we only have to show $\forall A, R, T \text{ Nat } R \cong R^n T$

By parametricity, $M \sim M$ define $\sigma(x) := A$ $\sigma'(x) := \text{Nat}$
 $\eta(x) := \bigcup_{n \in \mathbb{N}} [R^n T]_{\sim} [\underline{n}]$

We have $M \text{ Nat } R \sim M \text{ Nat } S \subseteq [\eta : \sigma \mapsto \sigma']$

since $T \sim_x \subseteq [\eta : \sigma \mapsto \sigma']$ with $n=0$

and $R \sim_{x \mapsto x} S [\eta : \sigma \mapsto \sigma']$

as $\forall U \sim_x V [\eta]$, then $U \cong R^n T$ and $V \cong \underline{n}$ for some n .

and thus $RU \cong R^{n+1} T$ and $SV \cong S\underline{n} = \underline{n+1}$

therefore $RU \sim_x SV [\eta]$.

And so we conclude.

Exercise 74.

By ext, it suffices to show that

$$\forall M, \text{ Succ}_1 M \cong \text{Succ}_2 M$$

By ex 73, $M \cong \underline{n}$ for some $n \in \mathbb{N}$,

and thus $\text{Succ}_1 M \cong \text{Succ}_1 \underline{n}$ by context closure

$$\text{Succ}_2 M \cong \text{Succ}_2 \underline{n}$$

and so we conclude.